

Quantitative Reasoning

Formula Sheet: Sets, Permutation, Combination, Probability

1. SETS: Principle of Inclusion and Exclusion

Notation: $n(A)$ = number of elements in A | $A \cup B$ = A ya B (union) | $A \cap B$ = A aur B dono (intersection)

Two sets

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

- Disjoint sets (kuch common nahi): $n(A \cap B) = 0$, to $n(A \cup B) = n(A) + n(B)$
- Sirf A (B ke bina): $n(A) - n(A \cap B)$
- Na A na B: $n(U) - n(A \cup B)$
- Complement: $n(A') = n(U) - n(A)$

Three sets

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

Yaad rakhne ka tareeka: Singles jama $\rightarrow$ Pairs minus $\rightarrow$ Triple plus

2. PERMUTATION (Tarteel / Arrangement)

Kab use karein: jab **order matter** karta ho (seating, rank, password, letters arrange karna).

Basic formulas

Situation	Formula
Factorial	$n! = n \times (n-1) \times \dots \times 2 \times 1$, $0! = 1$
n mein se r arrange (repetition nahi)	${}^n P_r = n! / (n-r)!$
Saari n cheezein arrange	$n!$
Repetition allowed	n^r
Kuch cheezein same hon (word arrangement)	$n! / (p! \times q! \times r!)$ (p, q, r = repeated cheezon ki ginti)
Circular arrangement	$(n-1)!$

Conditions: kab kya karna hai

Condition	Kya karein
Kuch cheezein hamesha saath hon	Unko ek unit (block) maano, baaki ke saath arrange karo, phir block ke andar ke arrangements se multiply karo. Misaal: n cheezein, 2 saath $\rightarrow (n-1)! \times 2!$
Do cheezein kabhi saath na hon	Total arrangements – (saath wale arrangements)
Kisi position par cheez fixed ho	Usko nikaal do, baaki $(n-1)$ arrange karo $\rightarrow (n-1)!$
Vowels saath hon (word)	Vowels ka ek block banao, block ke andar bhi arrange karo aur multiply karo
Number banana, zero se start na ho	Pehli jagah ke liye non-zero choices, phir baaki positions fill karo
Even / odd number	Pehle last digit (unit place) fix karo, phir baaki positions

Keywords: arrange, order, rank, seat, password, code $\rightarrow$ Permutation.

3. COMBINATION (Selection / Chunnna)

Kab use karein: jab **order matter nahi** karta (committee, team, cards select karna).

$${}^nC_r = n! / [r! (n - r)!]$$

Properties

- ${}^nC_r = {}^nC_{n-r}$
- ${}^nC_0 = {}^nC_n = 1$ | ${}^nC_1 = n$
- ${}^nP_r = {}^nC_r \times r!$

Conditions

Condition	Formula / Rule
Alag groups se select (m mein se a, n mein se b)	${}^mC_a \times {}^nC_b$ ("aur" → multiply)
Ya is group se ya us group se	Cases ko add karo ("ya" → add)
Kam az kam 1 select karna	Total – (koi bhi select na ho)
Specific cheez hamesha shamil	${}^{n-1}C_{r-1}$
Specific cheez kabhi shamil na	${}^{n-1}C_r$

Permutation vs Combination

	Permutation	Combination
Order	Matter karta hai	Matter nahi karta
Keywords	Arrange, rank, seat, code	Select, choose, committee, team
Formula	$n! / (n-r)!$	$n! / [r!(n-r)!]$
Value	Bari	Choti (r! se divide)

4. PROBABILITY

$$P(E) = \text{Favourable outcomes} / \text{Total outcomes} = n(E) / n(S)$$

- **Sample space (S)** = saare mumkin outcomes; **Event (E)** = jo outcome chahiye
- **Complement:** $P(E') = 1 - P(E)$, $P(E) + P(E') = 1$

Common sample spaces

- Coins: 1 coin = 2, 2 coins = 4, 3 coins = 8, **n coins = 2^n**
- Dice: 1 die = 6, 2 dice = 36, **n dice = 6^n**
- Cards: 52 total (26 red, 26 black; 13 each of spades, hearts, diamonds, clubs; 4 kings, 4 queens, 4 jacks; 12 face cards)

Rules

Rule	Formula
Addition ("ya" / OR)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Mutually exclusive (dono saath nahi ho sakte)	$P(A \cup B) = P(A) + P(B)$
Multiplication, independent ("aur" / AND)	$P(A \cap B) = P(A) \times P(B)$
Multiplication, dependent	$P(A \cap B) = P(A) \times P(B A)$
Conditional probability	$P(B A) = P(A \cap B) / P(A)$
Combination ke saath	$P = (\text{favourable ways}) / (\text{total ways})$, dono nC_r se. Misaal: 2 cards, dono king → ${}^4C_2 / {}^{52}C_2$

5. CONDITIONS OF PROBABILITY (Types of Events)

Event type	Probability	Matlab
Impossible	$P(E) = 0$	Kabhi nahi ho sakta (die par 7 aana)
Unlikely	$0 < P(E) < 0.5$	Hone ka chance kam
Equally likely / Even chance	$P(E) = 0.5$	Hone aur na hone ka barabar chance
Likely	$0.5 < P(E) < 1$	Hone ka chance zyada
Certain (Sure)	$P(E) = 1$	Zaroor hoga (die par 1 se 6 ke beech number)

Range: $0 \leq P(E) \leq 1$ (kabhi negative ya 1 se zyada nahi)

Number line: 0 (Impossible) — Unlikely — 0.5 (Even chance) — Likely — 1 (Certain)

Aur types

- **Mutually exclusive:** dono ek saath nahi ho sakte $\rightarrow P(A \cap B) = 0$
- **Independent:** ek ka dusre par asar nahi $\rightarrow P(A \cap B) = P(A) \cdot P(B)$
- **Exhaustive:** inki probabilities ka total = 1
- **Complementary:** E aur E' $\rightarrow P(E') = 1 - P(E)$